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The Relationship of Male Transsexual Typology to Psychosocial Adjustment

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In a search to uncover variables predictive of psychosocial adjustment of male transsexuals, this study examined features which have been empirically linked to characteristics of male transsexual typology. Data from 25 male transsexual volunteers were examined by means of multiple regression analysis to determine whether symptoms of introversion, depression, or tension, as well as adjustment to work and gender reorientation could be linked to the typological variables of androphilia, gynephilia, cross-gender fetishism, feminine gender identity in childhood, and age of onset of transsexualism. Valid and reliable scales measuring the typological variables of erotic partner preference (androphilia and gynephilia), cross-gender fantasy in association with sexual arousal (cross-gender fetishism), and degree of feminine gender identity in childhood were used. Results indicate a significant relationship between social gender reorientation and the feature of androphilia and between work adjustment and gynephilia. This differential adaptation is explained in terms of the different course that transsexualism takes within each typological subgroup. These findings offer evidence as to why some transsexuals may more readily adopt a female gender identity than others. There was no relationship found between the typological features and those variables measuring psychological disturbance.

KEY WORDS: transsexualism; cross-gender fetishism; cross-gender identity; erotic partner preference; gender reorientation.

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INTRODUCTION

One of the earliest debates in the literature on transsexualism concerns whether transsexuals are basically mentally sound or whether they suffer from psychopathological conflict. Research into this controversy began to accumulate in the early 1970s. The early results (Stinson, 1972; Finney *et al.*, 1975) suggested that, as a group, transsexuals showed clear signs of significant ego defect, or at least some degree of psychopathology. Other investigators (Rosen, 1974; Roback *et al.*, 1976; Tsushima and Wedding, 1979) indicated, however, that though some transsexuals may show a vulnerability to mental health adjustment, no real signs of psychopathology have been evidenced, especially when transsexuals are compared as a group to other psychiatric populations.

These results led some investigators to suspect that the research question was more complex and that the entire transsexual population could not be usefully defined in such dichotomous terms as psychologically sick or healthy. Langevin *et al.* (1977) then suggested that some factor within the syndrome may discriminate between transsexuals who are psychologically healthy or unhealthy, and they began to look at the amount of time each of their subjects had spent in the feminine role. They found that this factor seemed to discriminate clearly between adjusted and unadjusted groups. It was Greenberg and Laurence (1981), however, who actually clarified the relationship of living-in-the-role of a woman and level of disturbance. They tested their hypothesis on four patient groups and concluded that this sample characteristic accounted for the discrepancies in the levels of psychopathology demonstrated by the various groups in the studies previously cited.

As is typical in many types of research, the issue of debate evolved into a more complex question than it began with and the search for an answer to the question of whether transsexuals are inherently psychologically disturbed or well has shifted to an investigation of the variables that may predict or assist their mental adjustment.

One study which attempted to systematically identify variables which may moderate psychological adjustment in male transsexuals was conducted by Blanchard *et al.* (1983). They formalized the idea of living as a woman as a set of steps they call "gender reorientation" (p. 504). Basically, this variable has two components: a social component reflecting whether the transsexual is living as a woman at home, socially, and in the identification cards carried; and a physical component reflecting whether they have begun hormone treatment or have had genital surgery. These factors were compared by means of multiple regression analyses to two measures of psychological adjustment, the MMPI Tension and Depression Cluster Scales (Stein, 1968) and two measures for social adjustment, involvement and cohabitation. "Involvement" was defined as having been involved with a steady partner in the

past year, and "cohabitation" meant currently living with a lover or spouse. Results indicated a significant negative correlation with the measures of social feminization and the tension and depression scales and a significant positive correlation between genital surgery and cohabitation with a partner. Thus it was shown that social and physical gender reorientation appear to be highly correlated to several indices of mental health adjustment. However, making significant progress in undertaking the gender reorientation process only offers us another criterion to measure the psychological adjustment of transsexuals and offers no information about those who are poorly adjusted or those who may be having trouble undertaking the feminine reorientation process. For the proposed study, it seemed reasonable to assume that a more explanatory source for variables moderating mental health might come from the factors that describe consistent differences in the male transsexual populations; that is, the male transsexual typology.

The investigation of typological subgroups in the area of transsexualism dates from the early 1900s (Hirschfeld, 1910, 1922). Since then many others have added to this literature (American Psychiatric Association, 1980; Benjamin, 1966; Bentler, 1976; Buhrich and McConaghy, 1978, 1979; Meyer, 1974; Person and Ovesey, 1974a, 1974b; Stoller, 1968). Fortunately, there is widespread agreement on the categories of behavior that differentiate types. In general, the clinical features that reoccur consistently are those relating to erotic partner preference, presence or absence of cross-gender fetishism, and to a lesser degree, recalled feminine gender identity in childhood.

Further empirical support for this hierarchy of descriptors has been revealed by Freund *et al.* (1982). The principle component analysis of these and several other variables indicated that the dimension of heterosexuality versus homosexuality, cross-gender fetishism, and feminine gender identity in childhood accounted for 47% of the variance in subjects' scores on scales measuring these factors. These data not only support the clinically observed descriptions of transsexual types, but increase the evidence that there may be two separate etiological tracks that develop into transsexualism: one track proceeding from effeminate homosexuality and another from heterosexual cross-gender fetishism (Freund *et al.*, 1982; Buhrich and McConaghy, 1977, 1978, 1979).

In addition to the four established typological variables, age of onset was also selected as a variable that may be important in relation to both typology and etiological theory. Though often recorded for the purposes of clinical diagnosis, age of onset does not appear to have been systematically studied. For the purposes of this study, this variable was defined as the age at which evidence of cross-gender behavior or fantasy begins to appear. Evidence that this variable may be related to typology is implied from research on cross-gender-identified children (Green, 1987). These data suggest that overt feminine behavior in boyhood is more strongly associated with homosexuality and con-

sequently homosexual transsexuals. If this is true, then the age at which the earliest memory of these behaviors is recalled should be younger for the homosexual transsexual than for the heterosexual fetishistic group, and this factor should be related negatively to homosexual transsexualism. That late onset may be related to psychopathology is suggested from an epidemiological study by O'Gorman (1982). He found that transsexuals who showed a later onset experienced enuresis in childhood and had a history of psychiatric illness.

This study, then, proposed to examine the variables that have been established as most descriptive of the transsexual syndrome in relation to indices of psychoscial adjustment and disturbance. A significant relationship between any of these variables and mental health may suggest, at least, whether there are any psychosexual or developmental factors that can predict more consistently the psychological adjustment or disturbance of male transsexuals.

METHOD

Subjects

The subjects were 25 volunteer males recruited from the Seattle community. The principal source for the participants was the Ingersoll Gender Center, a nonprofit agency organized for the purpose of providing a support group and counseling referral services to individuals with gender conflicts. The main recruiting strategy was an advertisement given out through support groups, therapists, and an electrologist serving the transsexual community.

Participants were screened on the basis of their feminine gender stability. This was assessed through the reponse to the question,"Have you ever felt like a woman?" Those responding "at all times for at least one year" were included in the sample. This question has been used by others (Freund *et al.*, 1982; Blanchard, 1985) and found to be reliable when compared to other clinical screening methods. In addition, the majority of the respondents had been in the transsexual community for at least 4 to 5 months and were known to group leaders and others to be serious about pursuing gender change. All participants indicated they either planned to have genital surgery, or were postsurgical. No one was judged to be psychotic. All volunteers were paid $20 for completed questionnaires.

Materials

The instruments are grouped into two categories, predictor variables designed to measure the various typological features of male transsexualism and criterion variables chosen to measure specific dimensions of psychological, social, and psychogenderal functioning. The predictor variables erotic

partner preference were measured by the Modified Androphilia Scale (Andro) and the Modified Gynephilia Scale (Gyne) (Freund *et al.,* 1982; Blanchard, 1985). Androphilia (or androphile) is a term that refers specifically to a homosexual male who prefers a physically mature partner as opposed to a younger, immature sex partner, and a gynephile is the opposite or a heterosexual male who prefers a physically mature sex partner. Cross-gender fetishism was measured by the Cross-Gender Fetishism Scale (Blanchard, 1985). Cross-gender identity in childhood was measured by a version of the Feminine Gender Identity Scale (A) Revised (FGIS) (Freund *et al.,* 1974, 1977). Blanchard (1985) recommended several changes in this scale to increase its validity as a measure of femininity in childhood. Incorporating these changes resulted in a 16-item scale with all items referenced to childhood (ages 6–12) and a few to adolescence. Age of onset (Onset) was measured by two questions: "About what age were you when you first recall wishing you were a girl or woman?," and "About what age were you when you first dressed yourself in girls' or women's clothing?" The age of onset was determined by the earliest of the two reported ages, if there was a 4 or more year range between the two, or if the ages were within a 4-year period, they were averaged.

The criterion variables measuring psychological disturbance were three MMPI Cluster Scales, which are part of a group of seven scales developed by Stein (1968) to overcome the problem of overlapping items in the MMPI. The three scales chosen are TSC I, social introversion (Introv); TSC IV, depression and apathy (Depr); and TSC VII, tension, worry, and fear (Tens).

As described in the literature review, a form of social adjustment demonstrated specifically by transsexuals is gender reorientation. In this study, the instruments used to measure this criterion variable are the same as the social feminization and physical feminization measures used by Blanchard *et al.* (1983) with the following changes: The third item of the social feminization variable (GR Soc) substitutes "in your name on your driver's license" for "in your name on your documents"; and physical feminization (GR Phy) combines "hormones" and "vaginoplasty" into one variable.

To broaden the scope of the behavioral criterion, a variable measuring work adjustment (Work Adj) was also used. This single item variable asks, "Which of the following describes your current source of income or support for your living expenses?" The response, "full-time work" was scored 2 points; "part-time work," "savings," "retirement pension," and "student loans, grants, or scholarships" were scored 1 point. All other sources, "welfare," "parent," "spouse" were scored 0.

Procedure

Participants were asked to complete a 154-item questionnaire containing the questions designed to measure the predictor and criterion variables. They

came to the Ingersoll Gender Center to complete the questionnaire. In two cases, the questionnaire was sent to the participants and returned within 5 days.

RESULTS

Means and standard deviations for all study variables on 25 cases are shown in Table I. The data were analyzed by the SPSS New Regression subprogram (Hull and Nie, 1981) using the standard regression method (Nie *et al.*, 1975). Each of the six criterion variables were regressed separately against the five predictor variables. Standardized multiple regression coeffients (beta), explained variance (R^2), and significance of R^2 are summarized in Table II for all six analyses. The relationship between a predictor and criterion variable was considered significant if the probability of its occurrence in the data by chance was < 0.05. A Pearson product-moment correlation matrix (Table III) was also computed for all variables in the study and inspected to assure that the hypothesized relationships among the predictor variables held for this sample size and to verify that the criterion variables performed in the expected manner.

There was a statistically significant positive association between social gender reorientation (GR Soc) and androphilia (Andro) ($p > 0.0004$) and a trend ($p < 0.06$) for physical gender reorientation (GR Phy) to be related to Andro. There was also a tendency for gynephilia (Gyne) to be related to work adjustment (Work Adj) ($p < .07$). However, an inspection of the zero-order correlation matrix showed intercorrelational conditions (Pedhazur,

Table I. Means and Standard Deviations for all Study Variables on 25 Cases

Variable	$\bar{X}$	SD
Age	33.4	9.5
Educ	6.4	2.7
Work Adj	1.7	0.7
GR Soc	1.7	1.3
GR Phy	0.8	0.7
Introv	12.8	6.5
Depr	8.8	6.3
Tens	11.2	7.0
Andro	0.6	8.8
Gyne	2.0	5.5
CFS	0.008	7.4
FGIS	16.5	4.3
Onset	8.1	3.2

Table II. Standard Regression Analysis of Six Criterion Variables on Five Predictor Variables: Summary of Beta, R^2, and Significance of R^2

	Andro	Gyne	CFS	FGIS	Onset
Work Adj					
Beta	−.29304	.40458	−.29476	−.1330	.04554
R^2	.07818	.13114	.0638	.01256	.00163
Sig. R^2	.1631	.0757	.1907	.5677	.836
GR Soc					
Beta	.72033	.16025	.28001	.27851	.31386
R^2	.47242	.02057	.06107	.05483	.07743
Sig. R^2	.0004	.3827	.1382	.1609	.0992
GR Phy					
Beta	.41072	−.05728	.23314	−.17032	−.05378
R^2	.15358	.00263	.04278	.02050	.00227
Sig. R^2	.0653	.8008	.3149	.4835	.8144
Introv					
Beta	−.30573	−.29235	.09811	.03465	−.12458
R^2	.08510	.06847	.00757	.00085	.01220
Sig. R^2	.1802	.2272	.6827	.8910	.6045
Depr					
Beta	−.24246	−.25855	−.16316	.01774	−.20569
R^2	.05352	.05356	.02095	.00022	.03325
Sig. R^2	.2807	.2806	.4957	.9437	.3924
Tens					
Beta	−.00173	−.07519	−.12200	.14715	−.17032
R^2	.00000	.00453	.01171	.01530	.02280
Sig. R^2	.9940	.7594	.6230	.5746	.4942

1982) were present for cross-gender fetishism (CFS) to act as a suppression variable on gynephilia, suggesting that overlapping variance between these two predictor variables should be partialed out of the relationship between gynephilia and work adjustment. Accordingly, a hierarchical multiple regression between work adjustment and gynephilia with cross-gender fetishism as a control variable was performed (Table IV). The results indicate that a more realistic assessment of gynephilia's relationship to work adjustment is higher and significant ($p < 0.03$). None of the typological variables showed any significant relationship to the variables measuring psychological disturbance.

Some of the theorized relationships among the predictor variables were also not verified. The association between high scores on androphilia and high scores on feminine gender identity in childhood (FGIS) found in Freund *et al.*'s (1982) study was not replicated ($r = -.03$). In addition, the theorized relationship between Andro and earlier age for reported onset was also not found ($r = -.18$). Age of onset (Onset) did, however, show a significant negative relationship to FGIS ($r = -.34$) indicating as theorized that individuals with a greater feminine gender identity in childhood would show lower

Table III. Zero-Order Correlation Matrix for All Variables[a]

	Age	Educ	Work Adj	GR Soc	GR Phy	Intro	Depr	Tens	Andro	Gyne	CFS	FGIS
Educ	.4479 (.012)	–										
Work Adj	.0937 (.328)	.2197 (.146)	–									
GR Soc	−.0275 (.365)	−.3412 (.048)[b]	−.3115 (.065)	–								
GR Phy	.1937 (.177)	−.1356 (.259)	−.3136 (.063)	.6731 (.001)[c]	–							
Introv	−.2748 (.092)	−.0343 (.435)	−.0615 (.385)	−.2825 (.086)	−.3452 (.045)[b]	–						
Depr	−.2916 (.079)	−.1697 (.209)	−.0306 (.442)	−.3667 (.36)[b]	−.3422 (.047)[b]	.7944 (.001)[c]	–					
Tens	−.3164 (.062)	−.3501 (.043)[b]	−.1288 (.270)	−.0624 (.383)	−.0815 (.349)	.6513) (.001)[c]	.8246 (.001)[c]	–				
Andro	−.2298 (.135)	−.2161 (.150)	−.3054 (.069)	.6039 (.001)[c]	.3814 (.030)[b]	−.2622 (.103)	−.1436 (.247)	.0654 (.378)	–			
Gyne	.5400 (.003)[c]	.4011 (.023)[b]	.3767 (.032)[b]	.0694 (.371)	.0319 (.440)	−.2257 (.139)	−.2846 (.084)	−.1593 (.223)	−.1240 (.277)	–		
CFS	.3022 (.071)	−.0956 (.325)	−.0632 (.382)	.1065 (.306)	.1963 (.173)	.0454 (.415)	−.2036 (.164)	−.1797 (.195)	−.1730 (.204)	.3574 (.040)[b]	–	
FGIS	−.0175 (.467)	−.3532 (.042)[b]	−.2018 (.167)	.0601 (.388)	−.1881 (.184)	.1319 (.265)	.2123 (.154)	.2667 (.099)	.0354 (.433)	−.3214 (.059)[b]	−.2976 (.074)	–
Onset	−.3202 (.059)[b]	.0243 (.454)	.1524 (.234)	.0570 (.393)	−.0856 (.342)	−.0762 (.359)	−.1450 (.245)	−.2092 (.158)	−.1848 (.188)	−.0385 (.428)	−.0743 (.362)	−.3475 (.044)[b]

[a] *ps* in parentheses.
[b] $p = 0.01$–0.05.
[c] $p = 0.001$–0.003.

Table IV. Hierarchical Regression of Work Adj on Gyne With CFS As a Control Variable

Step	Variable entered	R^2	R^2 change	Sig.	Beta
1	CFS	.0039	−.0039	.7641	−.0631
2	Gyne	.1867	.1827	.0368	.4577

recalled ages for the onset of transsexualism. An unexpected significant negative correlation was found between Gyne and FGIS ($r = -.32$). Thus, though androphiles do not show a strong tendency to recall more feminine behavior and preferences in childhood, the gynephiles do recall significantly fewer of these behaviors and preferences.

DISCUSSION

The major finding of the present investigation was that the typological variables androphilia and gynephilia, or erotic partner preference, do appear to have a significant relationship to the social adjustment variables social feminization and work adjustment. Thus, those who scored higher on the Androphilia Scale manifest better adjustment in relation to gender conflict by their comparative readiness to take on a feminine presentation or role. With a larger sample, it is also likely that the tendency for androphiles to be more physically feminized would become a significant association. On the other hand, high scores on the Gynephilia Scale showed positive adjustment by their ability to generate their own income through some stable income resource(s).

These systematic adjustment differences between the heterosexual and homosexual types appear to have been noted over the years in various clinical data (Person and Ovesey, 1974a, 1974b; Meyer, 1974; Roback *et al.*, 1984). The data suggest that homosexual transsexuals tend to have, or take on, more femininity earlier in life and that the heterosexual types tend to be older and have developed a more masculine career and life-style before they undertake gender change. In their national survey, Roback *et al.* (1984) found these same differences existed particularly between younger and older groups of transsexuals. To explain these differences, they theorized that gender conflict for the older, more heterosexual group, is less intensely experienced early in life and becomes more intense only after traditional solutions have failed.

To amplify this hypothesis further, it is suggested that for the heterosexual transsexuals, the erotic component of cross-gender fetishism may provide an early outlet for the "intensity" of the cross-gender conflict. This "outlet" of cross-dressing and experiencing erotic release may delay the need for an early gender resolution by providing intermittent relief from gender distress until this erotic component fades later in adult life. Without such

an outlet, the homosexual transsexual experiences an increasing need for conflict resolution and is motivated to take on feminization at an earlier age. These behavioral differences in erotic genderal functioning between the two transsexual types may also reflect differences in the causal mechanism underlying the syndrome.

One study, of course, cannot verify that the behavioral typology within the two subgroups is unrelated to psychopathological vulnerability. However, all the multiple regression analyses showed a low association between the typological features and psychological disturbance variables, even though clearly high and low profiles (high and low psychopathology) made up most of the sample. In addition to the low association, 8 of the possible 12 regression coefficients (betas) between the typological variables and the psychological variables were also negative. The exceptions were the association of feminine gender identity in childhood (FGIS) with all the psychological variables and the relation of cross-gender fetishism with introversion. These associations were low and positive. Given the evidence on differential adjustment between the heterosexual and homosexual subgroup presented above, one might suspect that these differences might leave one group more vulnerable than the other. However, results in this study suggest little association between typology and psychopathology or that differences in adjustment style necessarily create psychological vulnerability.

Age of onset, proposed as a fifth variable for the typological study, did not perform exactly as theorized. Contrary to expectation, the homosexual subgroup (androphiles) did not report significantly lower ages for transsexual onset. Instead, lower reported age of onset appears to vary as a function of the number of recalled feminine behaviors in childhood. Since the heterosexual subgroup (gynephiles) were most likely to recall fewer of these childhood behaviors, it might be expected that they would also report older ages for onset. This logic, however, was not supported by this study. Though it was found that gynephiles are most likely to score lower on the FGIS variable, they do not systematically report older ages of onset, nor do the androphiles always report younger ones. This finding is also supported by Roback *et al.*'s (1984) survey which found no differences between their groups in reported age of onset. These data suggest, then, that it is likely that the syndrome manifests itself in both groups around the same time, though subsequently a different course is taken due to differences in erotic-genderal functioning.

Limitations and Recommendations

Clearly, the most limiting factor in this study is the sample size. Though most of the intercorrelations discussed are significant, they are of a lower magnitude than would optimally be preferred and thus are unstable. Results,

then, must be accepted cautiously and validated by further studies with a larger sample. Since this is also one of the few systematic investigations undertaken with a population sample it is also possible that some variables may not perform in the same way that they might in a more restricted one.

Finally, since the typological variables under study account, at most, for less than 50% of the variance in any one of the dependent variables, it is suggested that further studies include variables outside the typological domain as predictors. Variables that measures the quality of family life and interactions among family members and partners should be of particular importance to include, as the object relations or experiential domain is proposed as one area that may causally contribute to transsexualism.

REFERENCES

American Psychiatric Association. (1980). *Diagnostic and Statistical Manual of Mental Disorders,* 3rd ed., APA, Washington, DC.

Benjamin, H. (1966). *The Transsexual Phenomenon,* Julian Press, New York.

Bentler, P. M. (1976). A typology of transsexualism: Gender identity theory and data. *Arch. Sex. Behav.* 5: 567-584.

Blanchard, R., Clemmesen, L. H., and Steiner, B. W. (1983). Gender reorientation and psychosocial adjustment in male-to-female transsexuals. *Arch. Sex. Behav.* 12: 503-509.

Blanchard, R. (1985). Typological study of gender disorders in males. In Steiner, B. W. (ed.), *Gender Dyshporia: Development, Research, Management,* Plenum Press, New York.

Buhrich, N., and McConaghy, N. (1977). The discrete syndromes of transvestism and transsexualism. *Arch. Sex. Behav.* 483-495.

Buhrich, N., and McConaghy, N. (1978). Two clinically discrete syndromes of transsexualism. *Br. J. Psychiat.* 133: 73-76.

Buhrich, N., and McConaghy, N. (1979). Three clinically discrete categories of fetishistic transvestism. *Arch. Sex. Behav.* 8: 151-157.

Finney, J. C., Bransdsma, J. M., Tondow, M., and Lemaistre, G. (1975). A study of transsexuals seeking gender reassignment. *Am. J. Psychiat.* 132: 962-964.

Freund, K. (1985). Cross-gender identity in a broader context. In Steiner, B. W. (ed.), *Gender Dysphoria: Development, Research, Management,* Plenum Press, New York.

Freund, K., Langevin, R., Satterberg, J., and Steiner, B. (1977). Extension of the gender identity scale for males. *Arch. Sex. Behav.* 6: 507-519.

Freund, K., Nagler, E., Langevin, R., Zajac, A., and Steiner, B. W. (1974). Measuring feminine gender identity in homosexual males.

Freund, K., Steiner, B. W., and Chan, S. (1982). Two types of cross gender identity. *Arch. Sex. Behav.* 11: 49-63.

Green, R. (1987). *The "Sissy Boy Syndrome" and the Development of Homosexuality,* Yale University Press, New Haven, CT.

Greenberg, R. P., and Lawrence, L. (1981). A comparison of the MMPI results for psychiatric patients and male applicants for transsexual surgery. *J. Nerv. Ment. Dis.* 169: 320-323.

Hirschfeld, M. (1910). *Die Transvestiten,* Alfred Pulvermacher, Berlin, Germany.

Hirschfeld, M. (1922). Der Transvetitismus. In *Sexualpathologie,* Marcus & Weber, Bonn, Germany.

Hull, C. H., Nie, N. H. (1981). *SPSS Update 7-9,* McGraw-Hill, New York.

Langevin, R., Paitich, D., and Steiner, B. (1977). The clinical profile of male transsexuals living as females vs. those living as males. *Arch. Sex. Behav.* 3: 527-558.

Meyer, J. (1974). Clinical variants among applicants for sex reassignments. *Arch. Sex. Behav.* 3: 527-558.

Nie, N. H., Hull, C. H., Jenkins, J. G., Steinbrenner, K., and Brent, D. H. (eds.). (1975). *Statistical Package for the Social Sciences,* 2nd ed., McGraw-Hill, New York.

O'Gorman, E. C. (1982). A retrospective study of epidemiological and clinical aspects of twenty-eight transsexual patients. *Arch. Sex. Behav.* 11: 231-236.

Pedhazur, E. J. (1982). *Multiple Regression in Behavioral Research,* Holt, Rinehart & Winston, New York.

Person, E., and Ovesey, L. (1974). The transsexual syndrome in males: I. Primary transsexualism. *Am. J. Psychother.* 28: 4-29.

Person, E., and Ovessey, L. (1974b). The transsexualism syndrome in males: II. Secondary transsexualism. *Am. J. Psychother.* 28: 174-193.

Roback, H. B., McKee, E., Webb, W., Abramowitz, C. V., and Abramowitz, S. L. (1976). Comparative psychiatric status of male applicants for sexual reassignment surgery, jejunoileal bypass surgery, and psychiatric outpatient treatment. *J. Sex Res.* 12: 315-320.

Roback, H. B., Schwartz Fellman, E., and Abramowitz, S. I. (1984). The mid-life male sex-change applicant: A multiclinic survey. *Arch. Sex. Behav.* 13: 141-153.

Rosen, A. C. (1974). Brief report of MMPI characteristics of sexual deviation. *Psychol. Rep.* 35: 73-75.

Stein, K. B. (1968). The TSC scales: The outcome of a cluster analysis of the 550 MMPI items. In McReynolds, P. (ed.), *Advances in Psychological Assessment,* Vol.1, Science and Behavior Books, Palo Alto, CA.

Stinson, B. (1972). A study of twelve applicants for transsexual surgery. *Ohio State Med. J.* 68: 245-249.

Stoller, R. (1968). *Sex and Gender,* Science House, New York, p. 6.

Tsushima, W. T., and Wedding, D. (1979). MMPI results of male candidates for transsexual surgery. *J. Pers. Assess.* 43: 385-387.